CHAPTER

6

Capillary Electrophoresis: Principles and Instrumentation



Capillary electrophoresis (CE) is the primary methodology used for separating and detecting short tandem repeat (STR) alleles in forensic DNA laboratories worldwide. This chapter will examine the general principles and components of injection, separation, and detection of STR alleles using CE (Butler et al. 2004). The various Applied Biosystems (ABI) Genetic Analyzers, which are the instruments used worldwide for STR typing, will be discussed. Additional information on CE instruments including troubleshooting tips will be available in the forthcoming volume *Advanced Topics in Forensic DNA Typing: Interpretation*. Readers needing a more basic understanding of DNA separation and detection are encouraged to review Chapter 9 in *Fundamentals of Forensic DNA Typing* (Butler 2010).

REQUIREMENTS FOR STR TYPING

To achieve reliable STR typing, three conditions must be met. First, ***spatial resolution*** is needed to separate STR alleles that may differ in size by a single nucleotide (see Appendix 1). Second, ***spectral resolution*** is needed to separate fluorescent dye colors from one another so that PCR products from loci labeled with different dyes can be resolved. Third, *DNA sizing precision* from run to run must be consistent enough so that samples can be related to allelic ladders that are run for calibration purposes (see Figure 5.7).

These requirements have been met with a variety of CE systems. Since its introduction in July 1995 by Applied Biosystems, the single-capillary ABI Prism 310 Genetic Analyzer has been a popular method for STR typing in forensic DNA laboratories. More recently a variety

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TABLE 6.1 Information on Genetic Analyzer (Capillary Electrophoresis) Instrumentation Available from Applied Biosystems (Foster City, CA). Until the 3500 Series, All Systems Used Argon Ion (Ar^+) Lasers with 488/514 nm Wavelengths for Fluorescence Excitation. Information Courtesy of Michelle S. Shepherd, Applied Biosystems, LIFE Technologies.

ABI Genetic Analyzer	Years Released for Human ID	Number of Capillaries	Laser	Polymer Delivery	Other Features
373 (gel system)	1992–2003	–	40 mW Ar+ (488/514 nm)	–	PMTs and color filter wheel for detection
377 (gel system)	1995–2006	–	40 mW Ar+ (488/514 nm)	–	CCD camera
310	1995	1	10 mW Ar+ (488/514 nm)	syringe	Mac operating system & Windows NT (later)
3100	2000–2005	16	25 mW Ar+ (488/514 nm)	syringe	
3100-Avant	2002–2007	4	25 mW Ar+ (488/514 nm)	syringe	
3130	2003–2011	4	25 mW Ar+ (488/514 nm)	pump	
3130xl	2003–2011	16	25 mW Ar+ (488/514 nm)	pump	
3500	2010	8	10–25 mW diode (505 nm)	new pump	110 V power; RFID-tagged reagents; normalization & 6-dye detection possible
3500xl	2010	24			
3700	2002–2003	96	25 mW Ar+ (488/514 nm)	cuvette-based	Split beam technology
3730	2005	48	25 mW Ar+ (488/514 nm)	pump	
3730xl	2005	96	25 mW Ar+ (488/514 nm)	pump	

of capillary array instruments have become available and forensic laboratories have used the 3100, 3130, 3700, 3730, and 3500 series instruments. Table 6.1 reviews features of the various models of ABI Genetic Analyzers along with the number of capillaries used, which relates to the instrument's throughput capabilities.

PRINCIPLES AND COMPONENTS OF CAPILLARY ELECTROPHORESIS

The primary elements of a basic CE instrument include a narrow glass capillary, two buffer vials, and two electrodes connected to a high-voltage power supply. CE systems also contain a laser excitation source, a fluorescence detector, an autosampler to hold the sample tubes or tray, and a computer to control the sample injection and detection (Figure 6.1). CE capillaries are made of glass and typically have an internal diameter of 50 μm (similar to the thickness of a human hair). ABI Genetic Analyzers typically use capillaries with 36cm or 50cm injection-to-detection distances. For some higher-resolution work, such as long-read DNA sequencing, 80cm capillaries may be used.

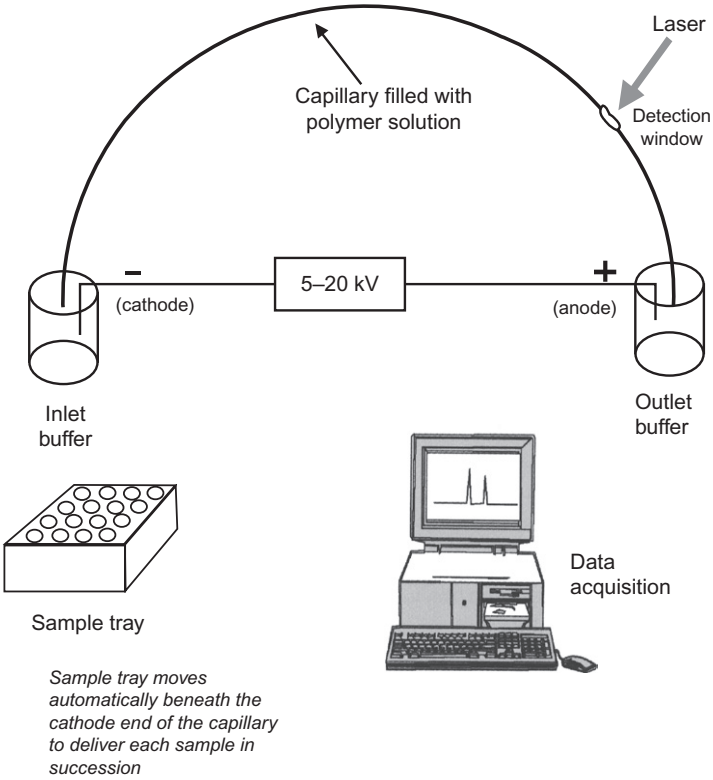


FIGURE 6.1 Schematic of capillary electrophoresis instruments used for DNA analysis. The capillary is a narrow glass tube approximately 50 cm long and 50 µm in diameter. It is filled with a viscous polymer solution that acts much like a gel in creating a sieving environment for DNA molecules. Samples are placed into a tray and injected onto the capillary by applying a voltage to each sample sequentially. A high voltage (e.g., 15,000 volts) is applied across the capillary after the injection in order to separate the DNA fragments in a matter of minutes. Fluorescent dye-labeled products are analyzed as they pass by the detection window and are excited by a laser beam. Computerized data acquisition enables rapid analysis and digital storage of separation results.

The same buffers that are used in gel electrophoresis may also be used with CE. However, instead of a gel matrix through which the DNA molecules pass, a viscous polymer solution serves as the sieving medium. The linear, flexible polymer chains act as obstacles that must be navigated by the negatively-charged DNA fragments on their way to the positive electrode. The larger DNA molecules move more slowly through the capillary than the smaller, more agile DNA fragments, which allows the molecules to be separated based on their size.

Prior to injecting each sample, a new gel is "poured" by filling the capillary with a fresh aliquot of the polymer solution. The CE capillary can be thought of as one lane in a gel that is only wide enough for one sample at a time. Different DNA samples are mixed with a constant set of DNA fragments of known size that serve as internal size standards in order to correlate results from run-to-run. An important difference between CE and gels is that the

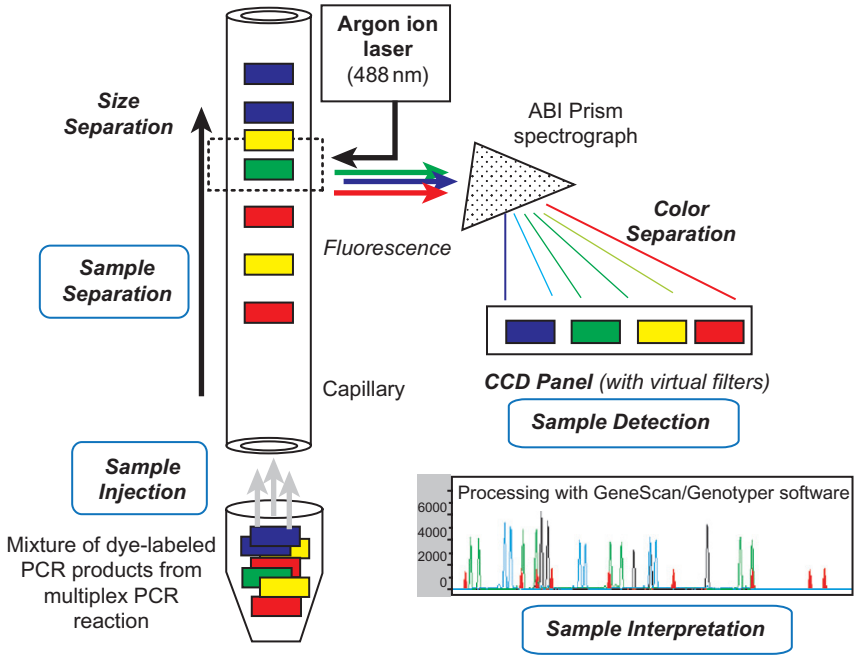


FIGURE 6.2 Schematic illustration of the separation and detection of STR alleles with an ABI Prism 310 Genetic Analyzer or a single capillary in an ABI 3100 or other multi-capillary instrument.

electric fields are on the order of 10 to 100 times stronger with CE (i.e., 300 V/cm instead of 10 V/cm), which results in much faster run times for CE.

Detection of the sample is performed automatically by the CE instrument by measuring the time span from sample injection to sample detection with a laser placed near the end of the capillary. Laser light is shined on to the capillary at a fixed position where a window has been burned into the coating of the capillary. DNA fragments are illuminated as they pass by this window. The smaller DNA molecules arrive at the detection point first followed by the larger molecules in order of their migration speed, which correlates with length or the number of base pairs. Data from CE separations are plotted as a function of the relative fluorescence intensity observed from fluorescence emission of dyes passing the detector. The fluorescent emission signals from dyes attached to the DNA molecules can then be used to detect and quantify the DNA molecules passing the detector (Figure 6.2).

ELECTROKINETIC INJECTION

The most common method used to introduce DNA samples into a capillary is a process known as *electrokinetic injection*, where a voltage is applied to a liquid sample immersed in one end of a capillary for a defined time. As DNA molecules are negatively charged in a neutral pH environment, a positive voltage draws the DNA molecules into the capillary.

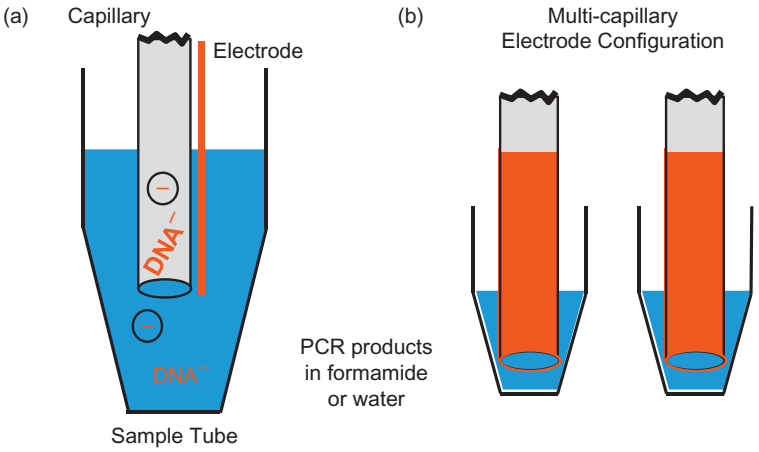


FIGURE 6.3 Comparison of capillary and electrode configurations with (a) single capillary ABI 310 and (b) multi-capillary systems. In (a) the electrode is adjacent to the capillary and in (b) the electrode surrounds the capillary.

Electrokinetic injections produce narrow injection zones that permit high-resolution DNA separations to occur in a relatively short separation distance.

In order to get DNA molecules onto the CE capillary, an electric voltage is applied while the end of the capillary is immersed into the liquid DNA sample. An electrode (cathode) is adjacent to the single capillary in the ABI 310 instrument while each capillary in capillary array instruments is enclosed in a platinum electrode (Figure 6.3). The flow of current generated by the voltage applied and the resistance experienced pulls the negatively-charged DNA molecules into the end of the capillary. However, CE injections of DNA are highly sensitive to the sample matrix—specifically the ionic strength of small negative ions, such as chloride from the PCR solution, provide competition for larger, slower DNA molecules in entering the end of the capillary.

The quantity of DNA injected into a CE column ($[DNA_{injected}]$) is a function of the voltage or electric field applied (E), the injection time (t), the concentration of DNA in the sample ($[DNA_{sample}]$), the area of the capillary opening (πr^2), and the ionic strength of the sample (λ_{sample}) versus the buffer (λ_{buffer}). The electrophoretic mobility (μ_{ep}), or how quickly a charged molecule moves when experiencing an electric field, also impacts the quantity of the DNA loaded into a capillary as does the electroosmotic flow (μ_{eof}), which will be explained further in the next section. The equation for amount of DNA injected—and hence detected—in a CE system is (Rose et al. 1988, Butler et al. 2004):

$$[DNA_{injected}] = \frac{Et(\pi r^2)(\mu_{ep} + \mu_{eof})[DNA_{sample}](\lambda_{buffer})}{\lambda_{sample}}$$

This equation reveals how a high sample ionic strength can impact the amount of DNA injected. Chloride (Cl^-) ions and other buffer ions present in PCR samples contribute to the sample conductivity and thus will compete with DNA for injection onto the capillary. This inverse relationship between sample conductivity (caused by its salt content) and amount of DNA injected into the capillary is the reason that post-PCR purification can increase the CE signal (see Chapter 11).

Sample Preparation

Sample preparation can impact electrokinetic injection. The PCR products created from amplifying a genomic DNA sample with an STR typing kit are typically diluted to levels of approximately 1 in 10 with deionized formamide (e.g., 1 µL PCR product into 9 µL of formamide) both to help denature the double-stranded DNA molecules and to help reduce the salt levels and aid the electrokinetic injection process.

Since formamide is a strong denaturant, it is commonly used in the preparation of single-stranded DNA samples for CE. Merely placing a sample in formamide is sufficient to denature it. However, heating a sample to 95ºC for several minutes followed by rapid cooling to around 4ºC (commonly referred to as "snap-cooling") is often performed to ensure that the two complementary DNA strands are separated or denatured.

Use of high-quality formamide with low conductivity is important. As formamide degrades it produces ionic decomposition products including formic acid, which are negatively charged at a neutral pH and will be preferentially injected into the capillary. The formamide by-products can cause problems in both sensitivity and resolution (Buel et al. 1998). The quality of formamide can be easily measured using a portable conductivity meter and should be approximately 100 µS/cm (the SI unit of conductivity is Siemens/meter) or lower to obtain the best results. The Hi-Di formamide (Applied Biosystems, Foster City, CA) used by most forensic DNA laboratories has a conductivity of <25 µS/cm and generally works well.

Many laboratories purchase ultrapure formamide and freeze aliquots immediately to ensure sample quality. Water has also been successfully used in the preparation of STR samples for CE analysis instead of formamide (Butler et al. 1995, Biega & Duceman 1999). Use of deionized water can eliminate the health hazard and the cost of formamide as well as problems associated with disposal. While studies have shown that water gives fully concordant results with formamide, long-term sample stability suffers because DNA molecules will renature (complementary strands will reanneal or come back together) in water after a few days.

The injection process moves the DNA molecules from their position in the sample tray onto the end of the capillary—essentially the starting line of a race where mixtures of different length molecules (amplified STR alleles) are separated.

DNA SEPARATION

Like marathon runners with different abilities, DNA molecules separate out along the race course of a capillary once a voltage is applied. Although the different length molecules begin the "race" together, the smaller ones move more quickly and are detected earlier at the "finish line." The time at which the DNA molecules cross the finish line is converted to base pair size through the use of internal size standards—analogous to runners with known abilities that move through the race course enabling a calibration of finish time with a runner's ability (DNA size).

There are several components that impact DNA separations within CE systems: the polymer used for enabling the separation, the capillary, the electrophoresis buffer, and the voltage applied or electric field strength (Butler 1995, Buel et al. 2003). STR allelic ladders, which contain multiple PCR products that are 4bp (or less) different in size, are useful tools for monitoring system resolution.

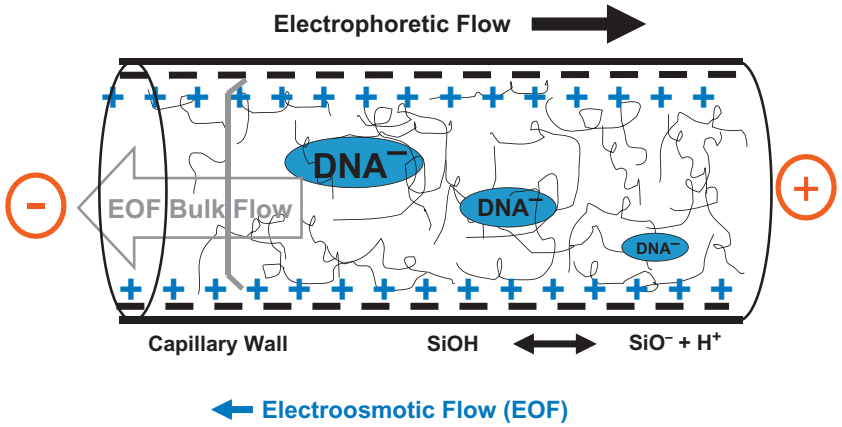


FIGURE 6.4 Illustration of internal capillary environment and forces at play with moving DNA molecules through a sieving polymer buffer.

Figure 6.4 illustrates the inner environment of a CE capillary. DNA moves with the electrophoretic flow from the negative electrode (cathode) shown on the left towards the positive electrode (anode) shown on the right. Strands of entangled polymer form transient pores that serve as obstacles to inhibit progress of DNA molecules based on their size. Smaller DNA molecules are able to move more easily through the obstructions.

As illustrated in Figure 6.4, another force is at play inside the capillary. The capillaries are made of glass or fused silica (silicon dioxide, SiO_2) but have hydroxyl groups (silanol, SiOH) along the inner walls and are negatively charged above approximately pH 5. Since the electrophoresis buffer used is typically around pH 8, positive ions from the buffer solution will line up with the negative siloxy (SiO^-) ions along the walls creating what is known as a double-layer. Under the electric field applied during CE separation (e.g., 15,000 volts), the positive ions will migrate toward the negative electrode. The movement of these positive ions will create a bulk flow of the solution within the capillary with what is known as *electroosmotic flow* or EOF. EOF can be reduced or eliminated by coating the inner capillary wall to prevent exposure of negative charges along the wall and buildup of positive ions from solution.

Polymer Solution

Applied Biosystems sells two primary polymer formulations for use with the ABI Prism 310 and other Genetic Analyzers: POP-4 and POP-6, where "POP" stands for Performance Optimized Polymer. These polymer solutions are 4% and 6% concentrations of linear, uncross-linked dimethyl polyacrylamide, respectively (Madabhushi et al. 1996, Madabhushi 1998). A high concentration of urea is also present in the polymer solution to help create an environment in the capillary that will keep the DNA molecules denatured.

POP-4 is commonly used for DNA fragment analysis including STR typing while the POP-6 polymer, which is the same polydimethylacrylamide polymer present at a higher

concentration, is capable of higher resolution to meet the single-base resolution needs of DNA sequencing. More recently, POP-7 polymer has been introduced for both STR typing and DNA sequencing applications.

Entangled polymers are characterized by a rapid increase in viscosity as the polymer concentration reaches a certain threshold value. The viscosity of these materials is also dependent on the polymer's relative molecular mass. The ideal polymer should have at least the same separation properties as classical polyacrylamide gels, combined with a low viscosity that allows easy replacement before each run is conducted on the CE instrument.

Buffer

The buffer that is used to dissolve the polymer in CE systems is important as it stabilizes and solubilizes the DNA and provides charge carriers for the electrophoretic current. If the buffer concentration and concomitant conductivity are too high, then the column will overheat resulting in a loss of resolution. In the process of electrophoresis, the composition of the anode and the cathode buffers may change due to electrolysis and migration of buffer ions. Thus, to avoid problems with poor size calibration of the system over time, it is a good policy to regularly replace the CE buffers with fresh solution.

The Genetic Analyzer buffer commonly used with the ABI 310 is 100 mmol/L TAPS and 1 mmol/L EDTA, adjusted to pH 8.0 with NaOH (Rosenblum et al. 1997). TAPS is short for *N*-tris-(hydroxymethyl)-methyl-3-aminopropane-sulfonic acid. TAPS is used instead of Tris-borate-EDTA (TBE) since TBE is temperature and pH-sensitive. As analysis temperature is increased with TBE, the pH decreases at a rate of 0.02 pH units with every 1ºC. As pH decreases so does the fluorescence emission of many dyes (Singer & Johnson 1997).

Under the analysis parameters typically employed for STR analysis, the amplified DNA fragments must remain denatured. To accomplish this DNA denaturation, the CE run temperature is set to higher than room temperature, and buffer additives such as formamide, urea, and 2-pyrrolidinone are added to keep the DNA from reannealing. Even under strong denaturing conditions, DNA molecules can sometimes assume various conformations due to intramolecular attractions. Therefore, capillary run temperatures of 60ºC are commonly employed to help reduce secondary structure in DNA (Rosenblum et al. 1997). In addition, high concentrations of urea and elevated temperatures are used to keep the various STR alleles uniformly denatured, since the mobility of DNA fragments can be affected by its conformation. Even with these measures, CE instruments need a stable ambient temperature, as temperature variations can have profound effects on allele migration (D.N.A. Box 6.1).

Many laboratories assess an internal standard peak (such as the 250 peak in the ABI GS500 internal standard), which is particularly sensitive to temperature variation to demonstrate that their CE systems are stable and well-calibrated (Klein et al. 2003). CE analysis of DNA fragments at elevated pH conditions, where the DNA molecule is predominately denatured, suggests that DNA secondary structure is responsible for the variations observed in DNA size determinations with fluctuating temperatures (Nock et al. 2001). By carefully controlling the run conditions, i.e., pH, buffer, denaturants, and temperature, variations within and between runs can be minimized and overall run precision improved. Run-to-run precision can also be enhanced using a global Southern sizing algorithm rather than the traditional local Southern sizing (Klein et al. 2003, Hartzell et al. 2003).

D.N.A. BOX 6.1

UNDERSTANDING CE INSTRUMENT SENSITIVITY TO ROOM TEMPERATURE CHANGES

Swings in room temperature of more than ≈2°C have been shown to adversely impact sizing precision when DNA separations are occurring in an ABI 310 or other Genetic Analyzer. This imprecision can lead to improperly calling "off-ladder" alleles when comparisons are made to sequentially-run allelic ladders. The reasons why temperature variation leads to imprecise measurements and inaccurate allele calls are enumerated below.

Voltage is determined by current times resistance ($V = I \times R$), which is also known as Ohm's law after Georg Ohm who discovered this relationship in 1827. Rearranging this equation, current is equal to voltage divided by resistance. Voltage is the energy put into the system by the power supply. In the case of an ABI 310 or 3100 Genetic Analyzer, the voltage delivered for the separation step is typically 15,000 or 15 kV. For the injection step, a lower voltage and hence lower current is often used to coax the DNA molecules into the end of the capillary. While thousands of volts may seem like a lot of energy, it is the current, or the flow of electrons or charged particles, that matters with electricity and electrophoresis. The speed at which the DNA molecules move is directly related to the current flow—not the voltage delivered. Resistance is high inside of a narrow glass capillary filled with a viscous polymer solution so current flow is typically in the 10 to 100 microampere region.

Excessive room temperature changes (swings of more than several degrees Celsius) can lead to temperature variation inside the capillary electrophoresis (CE) instrument and inside the capillary itself. Temperature changes that occur inside the capillary change the viscosity of the polymer which in turn alters the resistance experienced by the DNA molecules and other ions flowing during electrophoresis. As the polymer concentration inside the capillary effectively changes due to this temperature change, the mobility, or speed of movement, of DNA molecules shifts. If the temperature goes up, the polymer network of transient pores inside the capillary (see Figure 6.4) becomes more flexible (essentially less concentrated) and the DNA molecules move more quickly. Thus, while the voltage may be constant between runs, the resistance can change—leading to shifting in the current experienced by the DNA molecules undergoing electrophoresis.

Different length DNA sequences have different electrophoretic mobilities because they tumble or gyrate through a capillary or gel environment at different rates. Longer DNA molecules have a larger radius of gyration, which means that they will interact more frequently with polymer strands that effectively slow the DNA molecules and their movement through the capillary. Sequence differences in DNA molecules impact their 3-dimensional structure (their radius of gyration). Thus, in some cases, it is possible to separate two DNA molecules with the same length but different internal sequence (or to see a shift in the apparent size between them). This is the reason that allelic ladders are necessary in order to accurately genotype STR alleles separated and sized by electrophoresis. The allelic ladder alleles have the same DNA sequence as the STR allele being measured and thus possess the same radius of gyration

D.N.A. BOX 6.1 (*cont'd*)

when exposed to the same electrophoresis conditions.

How does this information relate to loss of precision in CE systems when room temperature varies? DNA sizes measured in CE are determined relative to an internal size standard, which contains a series of DNA fragments. These fragments have a different internal sequence compared to STR alleles. When the electrophoresis conditions change between two runs of the CE instrument (e.g., due to a room temperature shift that ultimately alters the polymer concentration in the capillary), the differential movement of the size standard peaks compared to the STR alleles may be sufficiently different between sequential runs to alter the apparent size of the STR alleles. If this happens, then the allelic ladder allele corresponding to the sample allele being measured may appear slightly different in size. If it falls outside of the ± 0.5 bp genotyping bin, then the sample allele is incorrectly labeled as "off-ladder." Thus, precise temperature control either within the instrument itself (or the room where the instrument is housed or both) is absolutely critical to obtaining quality STR typing results.

The Capillary

The capillary column is central to the separation capabilities of CE. As mentioned above, in uncoated capillary columns, residual charges on the silica surface induce a flow of the bulk solution toward the negative electrode. This EOF creates problems for reproducible DNA separations because the velocity of the DNA molecules can change from run to run. Capillary and microchip channel walls, which contain the charged siloxy groups, can be chemically modified (Hjertén 1985) or dynamically coated (Rosenblum et al. 1997) to prevent EOF in DNA separations. The charged sites on the wall can be masked by adsorption of neutral linear polymers that provide a viscous layer on the capillary surface.

The commercially available poly-dimethylacrylamide POP-4 and POP-6 polymers are successfully used in DNA genotyping by CE because they provide a sieving matrix for the separation of single-stranded DNA and, at the same time, suppress the EOF (Rosenblum et al. 1997). POP-4 consists of 4% linear dimethylacrylamide, 8 mol/L urea, 5% 2-pyrrolidinone (Rosenblum et al. 1997, Wenz et al. 1998).

Manufacturers of capillaries often suggest replacing a capillary at around 100 or 150 injections to avoid problems with resolution failure. Capillary lifetimes can be improved by rinsing the capillary with consecutive washes of water, tetrahydrofuran, hydrochloric acid, and polymer solution (Madabhushi 1998). Unfortunately, ABI CE systems do not permit an on-the-instrument wash so the capillary must first be removed to conduct the rinsing procedure. With good sample preparation, many forensic laboratories see capillary lifetimes extend far past the 100 injections recommended by the manufacturer. Through effective monitoring of sample resolution, capillaries can be replaced when resolution declines (Buel et al. 2003).

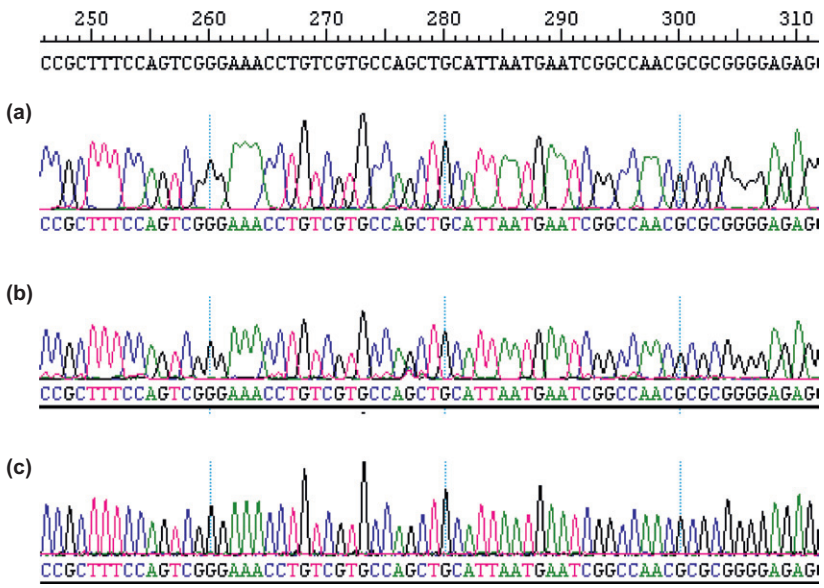


FIGURE 6.5 Impact of capillary length and polymer concentration on obtaining single base resolution in DNA sequencing. (a) POP-4 with 36 cm ABI 310 capillary (20 minute run time); (b) POP-6 with 50 cm ABI 310 capillary (120 minute run time); (c) POP-7 with 80 cm ABI 3130xl array (120 minute run time). Data courtesy of Tomohiro Takayama, NIST guest researcher.

Factors Impacting Resolution

A number of factors influence resolution in a CE system include capillary length, separation voltage, and polymer concentration. Separation of similarly-sized DNA molecules is achieved with an increased number of interactions with the entangled polymer strands while passing through the capillary. Thus, longer capillaries, lower separation voltages, and higher polymer concentrations improve resolution—but at the expense of longer separation times.

Figure 6.5 illustrates the benefit of a longer capillary and higher polymer concentration on the separation of DNA sequencing products that differ by a single nucleotide. A 36cm capillary with POP-4 polymer was barely able to subdivide the three "A" peaks at positions 262 to 264 (Figure 6.5a). These same three "A" peaks exhibited better resolution with a 50cm capillary and the more concentrated POP-6 polymer (Figure 6.5b). The best results, where baseline resolution was achieved, were seen with an 80cm capillary and POP-7 (Figure 6.5c). However, the better resolutions from the longer capillaries came at the expense of longer run times.

FLUORESCENCE DETECTION

Fluorescence-based detection assays are widely used in forensic laboratories due to their capabilities for multi-color analysis as well as rapid and easy-to-use formats. In the application to DNA typing with STR markers, a fluorescent dye is attached to a PCR

primer that is incorporated into the amplified target region of DNA. Amplified STR alleles are represented by peaks on an electropherogram as the labeled DNA molecules pass the detector (Figure 6.2). In this section, fluorescence detection is discussed in the context of the chemistry and instruments used to label and detect the PCR products produced from STR markers.

A significant advantage of fluorescent labeling over other methods is the ability to record two or more fluorophores separately using optical filters. The signal produced has to be spectrally resolved. This is accomplished using a fluorophore color separation algorithm known as a matrix, which is sometimes referred to as a "spectral calibration." With this multi-color capability, components of complex mixtures can be labeled individually and identified separately in the same sample.

A fluorescence detector is a photosensitive device that measures the light intensity emitted from a fluorophore. Detection of low-intensity light may be accomplished with a photomultiplier tube (PMT) or a charge-coupled device (CCD). The action of a photon striking the detector is converted to an electric signal. The strength of the resultant current is proportional to the intensity of the emitted light. This light intensity is typically reported in arbitrary units, such as relative fluorescence units (RFUs). A brief review of fluorescence is included in D.N.A. Box 6.2.

Optics for Genetic Analyzers

Multi-wavelength detection has expanded the capabilities of DNA analysis beyond a single dye color and permitted greater multiplexing for STR markers. The key to the utilization of this technology is to covalently bind a different dye onto the 5′ (nonreactive)-end of each primer or set of primers. These dyes have a number of interesting properties. They are all excited by a single argon ion laser tuned to 488 nm, yet fluoresce in different regions of the spectra.

Lasers are an effective excitation source because the light they emit is very intense and mostly at one or a few wavelengths. The argon ion gas laser (Ar^+) produces a blue light with dominant wavelengths of 488 nm and 514.5 nm. This laser is by far the most popular for applications involving fluorescent DNA labeling because a number of dyes are available that closely match its excitation capabilities. The ABI 3500 instrument utilizes a 505 nm diode laser.

The intensity of the light emitted by a fluorophore is directly dependent on the amount of light that the dye has absorbed. Thus, the excitation source is very important in the behavior of a fluorophore. Other important instrument parameters to be considered include optical filters used for signal discrimination and the sensitivity and spectral response of the detector.

A multi-wavelength analyzer, such as a CCD camera, can be used to determine which dye is present, based on the emission of each fragment as it passes the detector window. This technique permits the analysis of fragments of DNA that overlap in size as long as they are labeled with different dyes, which fluoresce at different wavelengths.

The ABI 310 Genetic Analyzer uses virtual filters to collect the light striking the CCD camera at particular wavelength intervals after the emitted light has gone through the optical path illustrated in Figure 6.6.

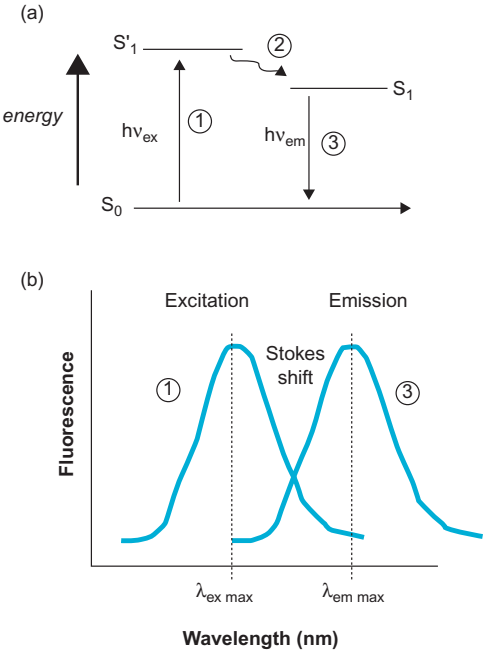
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FLUORESCENCE

Fluorescence measurements involve exciting a dye molecule and then detecting the light that is emitted from the excited dye. A molecule that is capable of fluorescence is called a *fluorophore*. Fluorophores come in a variety of shapes, sizes, and abilities. The ones that are primarily used in DNA labeling are dyes that fluoresce in the visible region of the spectrum, which consists of light emitted in the range of approximately 400 nm to 600 nm.

The fluorescence process is shown in the figure to the right. In the first step, a photon ($h\nu_{ex}$) from a laser source excites a fluorophore electron from its ground energy state (S_0) to an excited transition state (S'_1). This electron then undergoes conformational changes and interacts with its environment resulting in the relaxed singlet excitation state (S_1). During the final step of the process, a photon ($h\nu_{em}$) is emitted at a lower energy when the excited electron falls back to its ground state. Because energy and wavelength are inversely related to one another, the emission photon has a longer wavelength than the excitation photon.

The difference between the apex of the absorption and emission spectra is called the *Stokes shift*. This shift permits the use of optical filters to separate excitation light from emission light. Fluorophores have characteristic light absorption and emission patterns that are based upon their chemical structure and the environmental conditions. With careful selection and optical filters, fluorophores may be chosen with emission spectra that are resolvable from one another. This capability permits the use of multiple fluorophores to measure several different DNA molecules simultaneously. The rate at which samples can be processed is much greater with multiple fluorophores than measurements involving a single fluorophore.



There are a number of factors that affect how well a fluorophore will emit light, or fluoresce. These factors include the following:

- Molar extinction coefficient: the ability of a dye to absorb light;
- Quantum yield: the efficiency with which the excited fluorophore converts absorbed light to emitted light;
- Photo stability: the ability of a dye to undergo repeated cycles of excitation and emission without being destroyed in the excited state, or experiencing "photobleaching,' and;

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- Dye environment: factors that affect fluorescent yield include pH, temperature, solvent, and the presence of quenchers, such as hemoglobin.

The overall fluorescence efficiency of a dye molecule depends on a combination of these four factors.

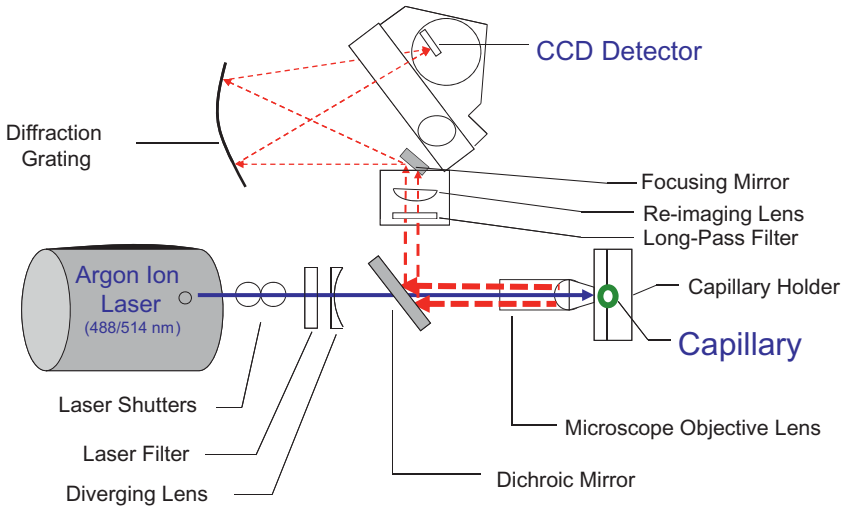


FIGURE 6.6 Optics in an ABI 310 Genetic Analyzer. Argon ion laser with 488 nm and 514.5 nm light (blue line) excites fluorophores passing through the capillary (green circle). Emitted light is projected through lenses, mirrors, and filters to a diffraction grating that spreads the wavelengths of light and then to a charged-coupled device (CCD) detector that simultaneously detects all wavelengths from 525 nm to 680 nm. Image adapted from Watts (1998). Figure courtesy of Bruce McCord, Florida International University.

Light from the argon ion laser passes through the laser filter, which removes low-intensity sidebands and other broadband spontaneous emission. Light is then focused by the diverging lens onto the dichroic mirror. The dichroic mirror is used to separate excitation and emission light paths. The beam then passes through the microscope objective to the sample. The emitted light is then reflected by the dichroic mirror onto a re-imaging lens after it passes through a long-pass filter.

The long-pass filter prevents light from the argon ion laser from interfering with the detection of the dye signals. Light is then directed onto a spectrograph where a diffraction grating disperses the light by wavelength and focuses the resulting spectrum onto a CCD array (Figure 6.6).

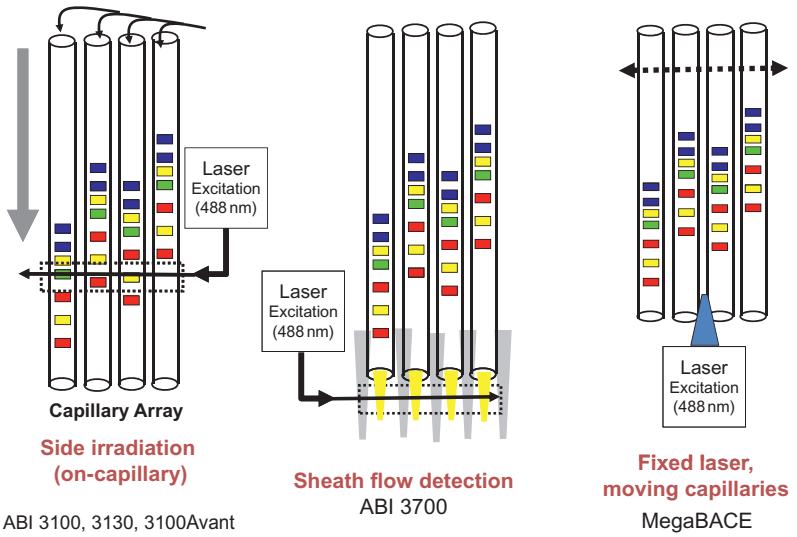


FIGURE 6.7 Different strategies for fluorescence excitation and detection with capillary array systems. The excitation beam is actually split and brought into both sides of the array with the 3100 and other ABI Genetic Analyzers.

With a single-capillary system like the ABI 310, the laser light can be trained continuously on the capillary enabling fluorescence excitation for any fluorophore passing the detector. However, with capillary array systems the laser light must illuminate multiple capillaries. Three approaches with capillary arrays have been utilized (Figure 6.7): on-capillary side irradiation, off-capillary sheath flow, and moving capillaries. The on-capillary side irradiation is used by the ABI 3100 and more recent capillary array systems from Applied Biosystems.

Fluorescent Dyes and Emission Filter Sets

Dye-labeled PCR primers label only a single strand of a PCR product. This simplifies data interpretation because the complementary DNA strand is not visible to the detector. In addition, dye-labeled primers enable multiple PCR products to be labeled simultaneously in an independent fashion thus increasing throughput capabilities because amplicons of overlapping size can be distinguished from one another by their dye label.

Figure 6.8a illustrates the range of fluorescence emission spectra for the different dyes commonly used to label PCR products in STR typing kits. The positions of light collected using several common virtual filters in the ABI 310 are noted in Figure 6.8b. The correct filter needs to be selected to match the fluorescent dye combinations in use in order to maximize sensitivity.

Spectral Calibration

In order for the computer software that is used for data analysis to know with what dye color a detected DNA fragment is labeled, the instrument detector and data collection

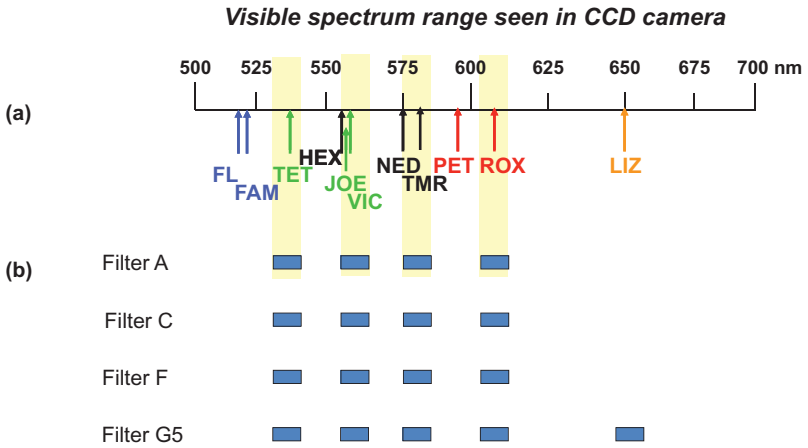


FIGURE 6.8 (a) Commonly used fluorescent dyes in STR typing kits shown by color with their spectral emission maxima (arrows). (b) Virtual filter sets used in the ABI 310 determine from which regions of the CCD camera visible light is collected for data interpretation.

software need to be calibrated. A multi-component spectral calibration is performed by testing a standard set of DNA fragments labeled with each individual dye, known as matrix standard samples. The term "matrix" comes from the use of multiple equations (e.g., samples labeled with a single dye) being used to solve multiple unknowns (e.g., the amount of fluorescent signal contribution from each dye color in other dye color channels) in the form of a mathematical matrix (e.g., 4×4 with four dye colors).

Computer software provided with the CE instrument then analyzes the data from each of the dyes and creates a matrix or spectral calibration file to reflect the color overlap between the various fluorescent dyes. These matrix files are a summary or template of how much overlap one should expect to see just by virtue of the dyes themselves given a particular instrument and environmental conditions. The difference between these matrix values and what is actually observed in the raw data becomes part of the data set.

Spectral calibration is unique to each instrument due to small variations in the laser and detector employed and other environmental conditions. This calibration needs to be performed on a regular basis as instrument and environmental conditions, such as laser excitation power, temperature, and solution pH, can drift over time and impact the fluorescence intensity of the dyes. A spectral calibration should be performed anytime a new laser or detector is installed. As long as the electrophoresis conditions are constant from run to run, then the emission spectra of the dyes should be reproducible and spectral overlap can be accurately deciphered.

If the matrix color deconvolution does not work properly to separate spectral overlap of the dyes used for STR allele detection, then the baseline analytical signal from the instrument can be uneven or a phenomenon known as "pull-up" can occur. *Pull-up* is the result of a color bleeding from one spectral channel into another, usually because of off-scale peaks.

The most common occurrence of pull-up involves small green peaks showing up under blue peaks when the blue peaks are several thousand RFUs (relative fluorescence units)

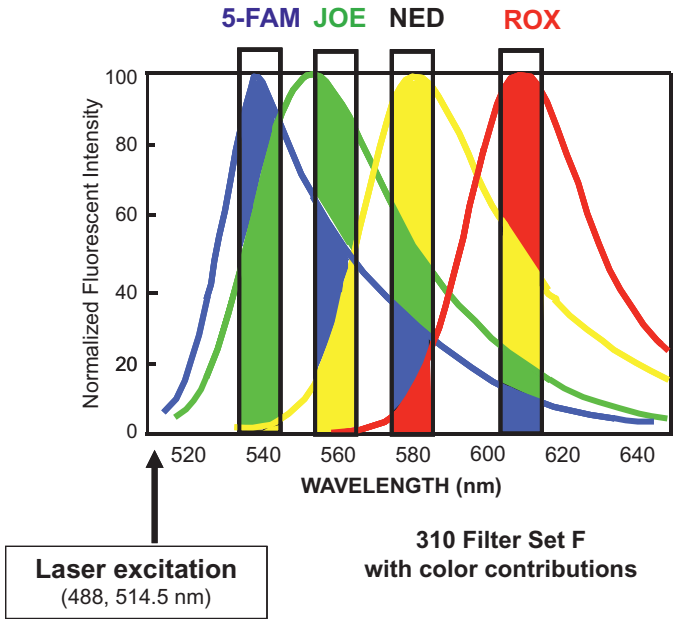


FIGURE 6.9 Fluorescent emission spectra of ABI dyes used with AmpFlSTR kits. ABI 310 Filter Set F is represented by the 4 boxes centered on each of the 4 dye spectra. Each dye filter contains color contributions from adjacent overlapping dyes that must be removed by a matrix deconvolution. The dyes are excited by an argon ion laser, which emits light at 488 and 514.5 nm.

in signal or off-scale. This occurs because of the significant overlap of the blue and green dyes seen in Figure 6.9. Samples can be diluted and analyzed again to reduce or eliminate the offending pull-up peak(s). Pull-up, also known as "bleed-through," is discussed in more detail the forthcoming *Advanced Topics in Forensic DNA: Interpretation* as it relates to data interpretation.

Raw data from a fluorescently labeled DNA sample is compared to the color-separated processed data in Figure 6.10. DNA fragments labeled with the yellow dye are commonly shown in black to make the display more visible.

The multi-capillary ABI 3100 series instruments automatically apply a spectral calibration to the CE separated fluorescence data as it is being collected so that there is no "raw data" generated by these instruments. Thus, if the color separation is not working properly, then new spectral calibration standards will have to be run and the previously poor performing samples (i.e., those exhibiting excessive pull-up) will have to be re-injected in order to obtain higher quality data.

Adjusting Dye Sensitivities with Variable Binning

Since not all of the dyes are equally sensitive when excited with a single laser wavelength, electronic signal enhancement has been used with newer versions of software to strengthen the weaker signals. ABI 3100 Data Collection software following the original v1.0.1 release

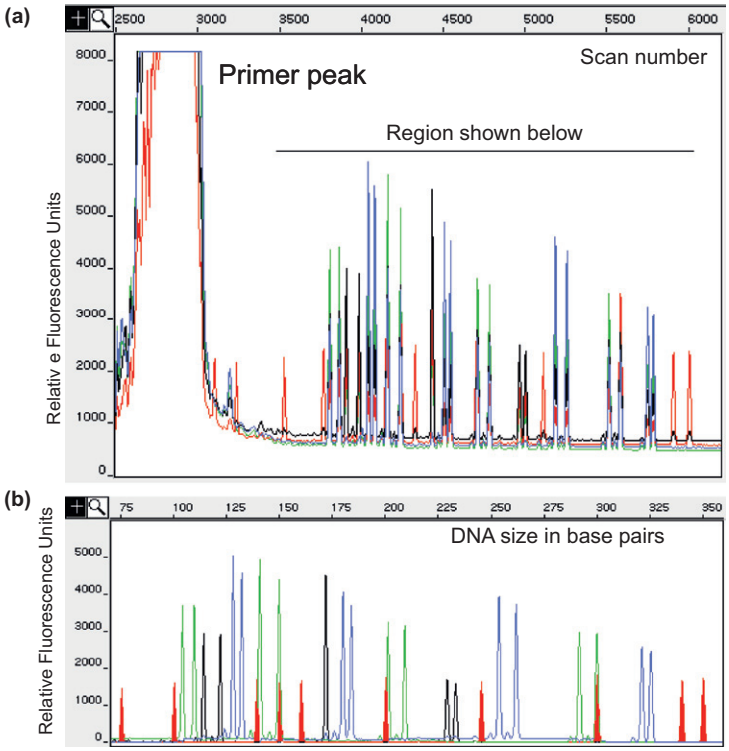


FIGURE 6.10 STR data from the ABI Prism 310 Genetic Analyzer. This sample was amplified with the AmpFlSTR SGM Plus kit. (a) Raw data prior to color separation is compared with (b) GeneScan 3.1 software color-separated allele peaks that have been sized using the internal size standard GS500-ROX (shown in red).

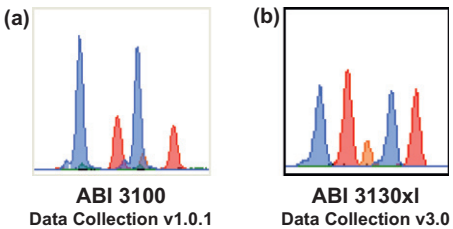


FIGURE 6.11 The same PCR products examined with different data collection versions. In (a) there is an equal number of pixels of light collected from the CCD camera for the blue-labeled and red-labeled peaks. In (b) the signal increase in the red dye-labeled PCR products is accomplished with "variable binning" where more pixels of light are collected from the CCD camera in the red channel to help balance the less sensitive red dye with blue dye-labeled amplicons.

(including versions 1.1, 2.0, and 3.0) tweak the red-color channels to balance the signal for 5-dye chemistry kits such as Identifiler (Figure 6.11). The technique is referred to as "variable binning" and involves collecting more light from the CCD camera with weaker dyes to help balance the overall signal output.

INSTRUMENTATION

A number of DNA separation and fluorescence detection platforms exist and have been used for STR allele determination (see Table 9.1 from the *Fundamentals* volume, Butler 2010). In the mid-to-late 1990s when STR typing first started, the gel-based ABI 373 and 377 instruments were the instrumental workhorses for both DNA typing and DNA sequencing (see Table 6.1).

Since the introduction of the single capillary ABI 310 in 1995 and a subsequent series of capillary array instruments including the ABI 3130xl, CE has now replaced slab gel systems throughout the forensic DNA community. CE systems are easy to use, automate data collection, and reduce labor compared to the early gel electrophoresis systems. With CE a set of samples are loaded into an instrument along with other reagents and the operator can walk away after starting the computerized data collection. There is no more need to clean gel plates, prepare polyacrylamide gels, or carefully load samples onto a gel with millimeter tolerances.

Overview of Sample Processing Steps

As has been described throughout this chapter, the single-capillary ABI 310 and multi-capillary CE systems perform DNA separations with multiple-color fluorescence detection and provide the capability of unattended operation. An operator prepares a batch of samples and then loads them in the instrument "autosampler," enters the name and well position in the autosampler tray for each sample into the data collection software, places a capillary and a syringe full of polymer solution in the instrument (or capillary array and bottle of polymer solution in the case of the ABI 3130xl), and starts the "run."

For the ABI 310, the data are serially processed at the rate of approximately one sample every 30 minutes of operation. Run times are usually slightly longer with the ABI 3100 or 3130 instruments (e.g., 35 to 45 minutes) but with the advantage of being able to collect information on either 4 or 16 samples simultaneously, depending on the number of capillaries present in the array. A major advantage of the technique for forensic laboratories is that the DNA sample is not fully consumed and may be retested if need be. In many cases, retesting is as simple as reinjecting the already prepared DNA samples.

The ABI Prism 310 Genetic Analyzer

Figure 6.12 shows the inside of an ABI 310 Genetic Analyzer. A single capillary is located between the gel block and the inlet electrode. The capillary is filled with polymer solution through the gel block. A heat plate is used to raise the temperature of the capillary to a specified temperature. Samples are placed in an autosampler tray that moves up and down to insert the sample onto the capillary and electrode for the injection process.

ABI 3100 and ABI 3130xl Instruments

The multi-capillary ABI 3100 Genetic Analyzer became available in 2000 and enabled higher throughputs compared to the single-capillary ABI 310 instrument. The 3130xl

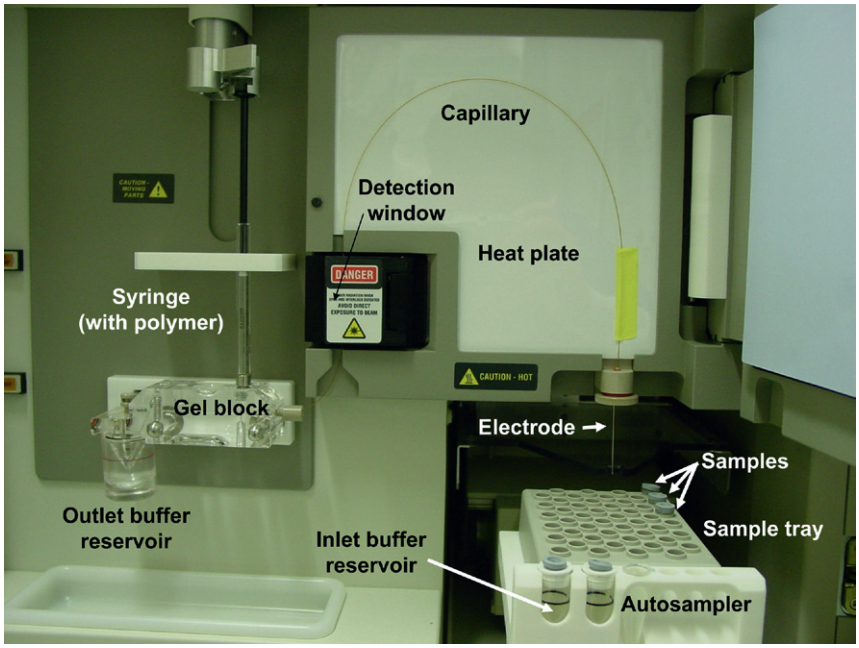


FIGURE 6.12 Photograph of ABI Prism 310 Genetic Analyzer with a single capillary.

16-capillary instrument became available in 2003 (Table 6.1). A photo of a capillary array as it appears inside the ABI 3130xl instrument is shown in Figure 6.13. The primary difference between the 3100 and 3130xl instruments is the use of a mechanical pump to fill the capillaries with polymer rather than a syringe as in the ABI 3100 (Figure 6.14).

Both 96-well and 384-well plates of samples may be processed in an ABI 3100 or 3130xl. With each run taking roughly 45 minutes, a 96-well plate can be analyzed in approximately five hours with six injections containing 16 samples each. Injections occur onto all 16 capillaries at once so if a user does not have 16 samples (2 columns of 8 samples in a 96-well plate), then formamide is put in the empty wells to prevent "dry" injections that can shorten the life of the capillaries.

An important difference between the ABI 310 and 3100 is that spatial and spectral calibrations are required prior to collecting data on the ABI 3100. The spatial calibration enables the CCD detector to know the location of each capillary while spectral calibration enables the dye colors to be resolved from one another.

Precision studies conducted on the ABI 3100 (Sgueglia et al. 2003, Butler et al. 2004) and the ABI 3700 (Gill et al. 2001) have demonstrated that reliable results can be obtained with multi-capillary CE systems.

ABI PRISM 3500 and 3500xl Genetic Analyzers

The ABI 3500 instrument (Figure 6.15) is available in 8 capillary (3500) or 24 capillary (3500xl) formats. This latest generation of CE instruments has several new features

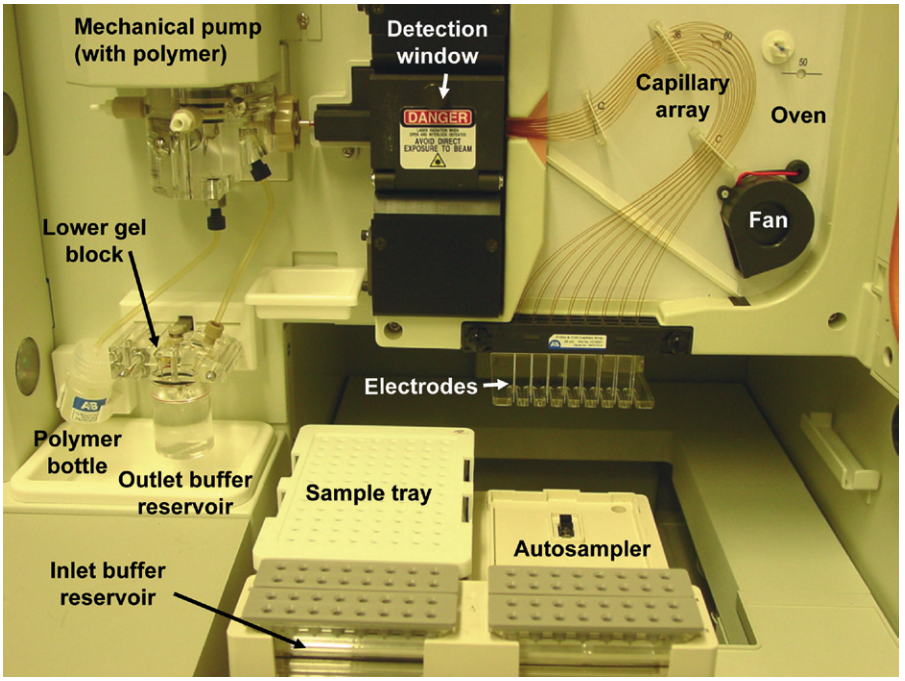
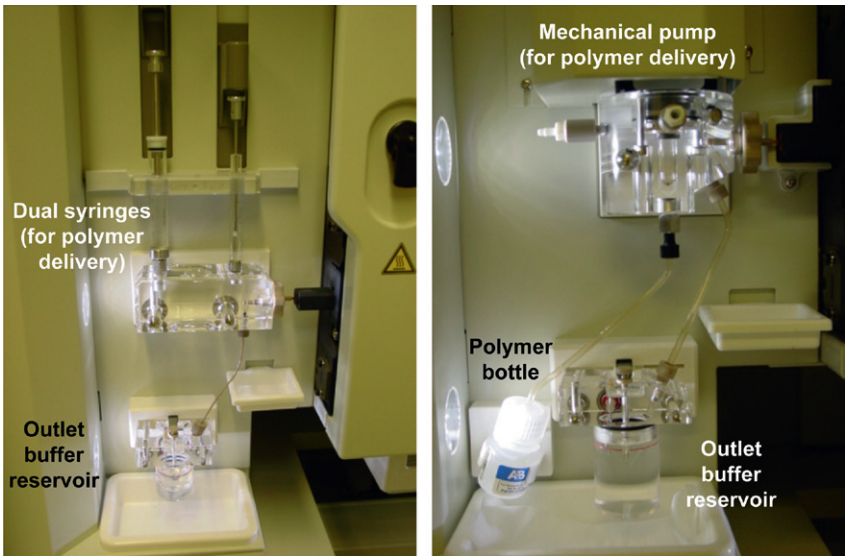


FIGURE 6.13 Photograph of ABI 3130xl Genetic Analyzer with a 16-capillary array.



(a) ABI 3100 **(b)** ABI 3130xl

FIGURE 6.14 (a) ABI 3100 syringe used to fill 16-capillary array. (b) ABI 3130xl mechanical pump used for polymer delivery from bottle.

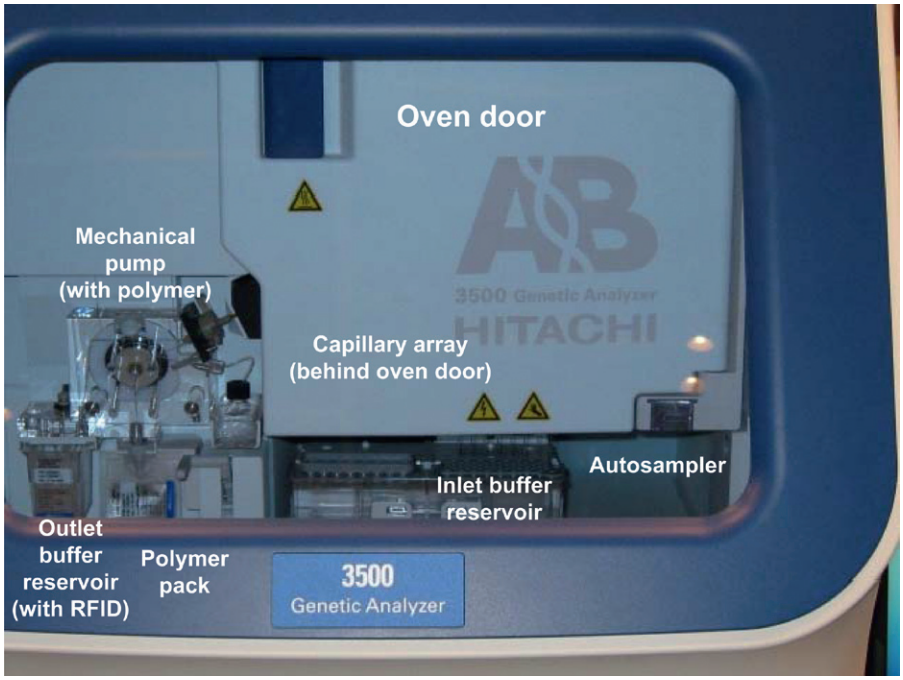


FIGURE 6.15 Photograph of ABI 3500 Genetic Analyzer (8-capillary array not shown).

including a smaller area for the array with better seals for more consistent temperature control, an improved mechanical pump for polymer filling of the capillaries with less waste, new laser technology (a solid state laser instead of an argon ion laser), reduced power requirements (110 V instead of 220 V), and 6-dye detection capability.

In addition, radio frequency identification (RFID) tags are used for tracking consumables (polymer, buffer, and capillaries) which are made available in ready-to-use formats to enable easier switching of reagents. The data collection software includes quality control features for rapid identification and re-injection of failed samples. With the use of a specific internal size standard (LIZ-600 v2), peak height normalization is possible leading to improved peak height uniformity across capillaries, runs, and instruments within a laboratory.

READING LIST AND INTERNET RESOURCES

Capillary Electrophoresis

Amplified DNA product separation. <http://dna.gov/training/separation/>.

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